BRIEF #13 · POWER FOR A CITY THE GRID CAN'T PROTECT

When the grid fails, the hospital runs on diesel.

Design and dispatch decisions are made blind — before the outage, before the money. Sentinel makes them computable.

7×

diesel cost per kWh vs Band A grid power — the economics every hospital director knows

12+

national grid collapses in 2024 (industry counts; NERC official count lower)

₦1,560/L

automotive gas oil, Lagos early 2026 (NBS price tracker)

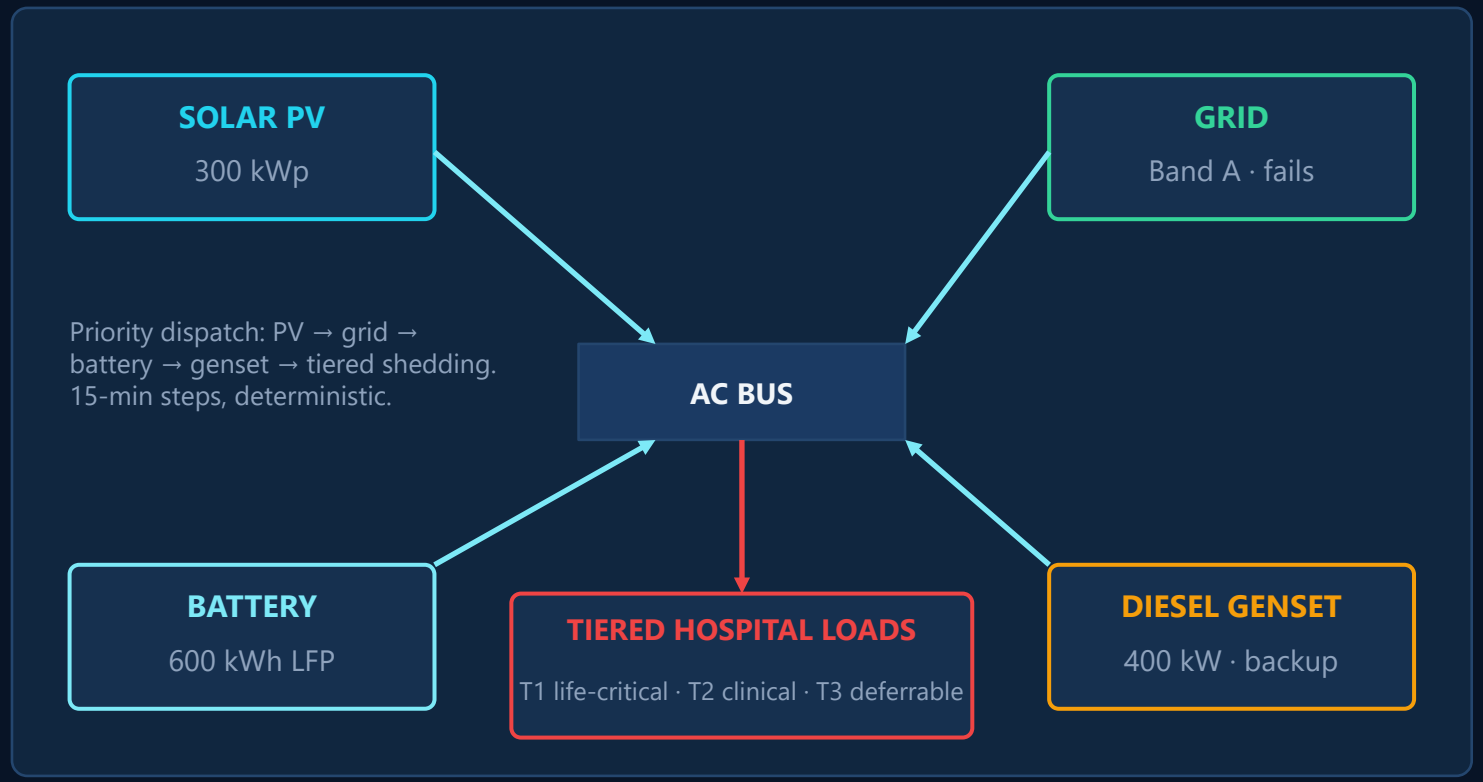
1,295 L

diesel burned by the typical backup in ONE 12-hour outage (engine baseline)

Sources: ICIR / APGC industry counts 2024–25 · NBS diesel price watch · Sentinel deterministic baseline simulation (12.5 h islanded window, 250-bed reference hospital).

The hospital microgrid, made predictable.

MedGrid Sentinel — a digital twin that designs, dispatches and prices the microgrid around the loads that keep patients alive, computed by a deterministic, unit-tested engine.



Deterministic core

15-min power-balance engine; 58 unit tests with hand-checked values; byte-identical replays

Scenario lab

Ten stress scenarios incl. the brief's line outage, generator failure and power-quality trips

Honest economics

EaaS fee vs savings with a computed breakeven financing rate — never an assumed grant

Deficits reported

When a design fails, Sentinel shows the failing hour — that is the product's value

Safety-critical dispatch is deterministic — no black box decides when the theatres keep power.

PROOF OF CONCEPT · BRIEF SCENARIO — 12.5-HOUR CITY LINE OUTAGE

Continuity held. Diesel cut 41%. Every number reproducible.

12.50 / 12.50 h

critical-service continuity (T1+T2) maintained through the outage — 100%

0.00 kWh

unserved Tier-1 life-critical energy — deficits reported, never hidden

−41%

diesel vs the diesel-only baseline in the 3-day window (763 L vs 1,295 L)

58 / 58

engine unit tests passing — physics, dispatch, finance, forecasting, verified build



The running application: islanded bus at 10:00 during the outage — generator carrying 197 kW after the battery bridge, event log showing the islanding sequence, zero synthetic alarms.

Source: Sentinel deterministic engine — regenerate with `npm run results` (results/demo-results.json). Illustrative facility profile; failure case (generator fault) reported at 98.2% continuity rather than hidden.

Annualized (320 normal + 45 outage days): 23,910 L/yr diesel avoided · 64.1 t CO_2/yr · ₦111.8M/yr operating savings

COMMERCIAL MODEL & FEASIBILITY · 6–18-MONTH PATHWAY

Honest economics, computed to the naira.

₦760.4M

CapEx — $491k for 300 kWp PV + 600 kWh LFP + dispatch on the existing genset

7.9 yr

simple payback against a 20–25 yr asset life; IRR 4.6%

₦161M vs ₦112M

annual EaaS fee vs operating savings at 14% financing

4.6%

financing rate at which the service closes — the computed concessional-capital gap

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Stated plainly: at an 18% commercial discount rate the project NPV is negative (−₦326M). The value case is resilience + diesel displacement + concessional capital — the structure real deployments like the NSIA-LUTH cancer-centre microgrid (300 kWp / 1.1 MWh, 2025) already use. No grant is assumed to exist anywhere in the model.

M 1–3

Metering-first pilot: real load + outage data on one feeder

→

M 4–6

Engineering design, permits, financing & O&M agreements

→

M 7–12

Pilot deployment; measured-vs-predicted evidence

→

M 13–18

Replicate to 3–5 facilities; provider playbook

Payer: the hospital (ownership) or an energy-service provider under EaaS · Maintainer: provider's local technicians · Sensitivity: diesel ±30%, CapEx ±20%, outage frequency 20–70 days/yr.

IMPACT · SCALABILITY · THE ASK

Built this weekend. Ready for a feeder near you.

23,910 L

diesel avoided per year (−41% outage-day; 45-day annualization)

64.1 t

CO_2 avoided per year — 2.68 kg CO_2 per litre, engine-computed

563 h

critical-service hours maintained per year across outage days

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Scales by configuration, not by code changes.

One facility = one config object: loads, tariffs, outage pattern, climate. The rural-clinic profile runs on the same engine today; the sizing optimizer already proves the design flips with outage timing (daytime → PV-first, evening → storage-heavy). Country adaptation = parameter change: tariffs, diesel/FX, standards, financing.

What we are not claiming: no measured facility data yet (inputs are labelled illustrative), no partners or pilots, no certifications — the 6–18-month plan is deliberately metering-first.

THE ASK

The 2027 Global Energy Cohort pathway: one pilot hospital, metered data, and a measured-vs-predicted proof pack within a year.

Team: Jackson Sixtus Abetianbe (solo entry) · all product features built during the 48-hour sprint · 58 tests · reproducible results.

MEDGRID SENTINEL — Intelligent Hospital Microgrids. Predictive Energy Resilience. Uninterrupted Critical Care.